

SIMATS ENGINEERING

Department of Computer Science and Engineering

ASSESSMENT TOOL 2

Industry Problem Solving Task

Designing Network Topologies Using Networking Devices,
Transmission Media, and Cabling Standards

Field	Details
Course Name	Computer Networks
Course Code	CSA07 (Subject Code: 0713)
Course Outcome	CO4 — Design network topologies (star, mesh, bus, hybrid) using networking devices, transmission media, and cabling standards (BL2, BL6)
Assessment Weightage	30%
Submission Type	Annexure A — Industry Problem Solving Task

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Code of Conduct — Declaration of Original Work

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Grading Rubric — Reference

The scoring rubric below is reproduced from the assessment brief for reference; the report is structured so that every criterion is demonstrably addressed (see Section 8 for the explicit mapping).

Criteria	Weight	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding
Solution Design	20%	Innovative, practical, industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, well-executed	Mostly functional, minor errors	Partially implemented	Incomplete or non-functional
Use of Tools & Technologies	10%	Advanced, appropriate use	Correct use of tools	Limited or basic use	Incorrect or no use
Problem-Solving Approach	10%	Logical, systematic, well-structured	Mostly logical approach	Somewhat disorganized	No clear approach
Innovation & Creativity	10%	Highly creative and original	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough, real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination	Good collaboration	Limited participation	No collaboration

1. Introduction

Computer network design is the process of translating an organisation's operational requirements — the number of users, the criticality of uptime, the physical layout of buildings, and the available budget — into a concrete arrangement of networking devices, transmission media, and cabling standards. No single network topology is universally optimal; the right choice always depends on the context in which the network must operate. A star topology that is perfectly adequate for a small office would be dangerously fragile for a bank, while the redundant mesh architecture that a bank requires would be needlessly expensive for a classroom.

This report addresses five distinct industry scenarios, each drawn from a different operating context, and for each one proposes and justifies a network topology, the supporting devices, the transmission media, and the cabling standards required to implement it. The scenarios covered are:

- A 20-person startup office, where ease of management and low cost dominate the decision.
- A bank with a near-zero downtime requirement, where fault tolerance and redundancy are paramount.
- A small classroom lab operating under a tight budget, where minimising hardware cost is the primary driver.
- A large university campus spanning multiple buildings, which requires a scalable, hierarchical hybrid design.
- A multi-department company network, where logical segmentation and inter-departmental routing are essential for security and performance.

Each section follows a consistent structure: a statement of the problem and its constraints, an analysis of the technical requirements, the proposed topology and justification, a labelled network diagram, a bill of devices and media, and a discussion of the design's strengths and limitations. A comparative analysis and a common testing and validation methodology are presented after the five scenarios, followed by an overall conclusion.

Course Outcome Alignment

This task is mapped to Course Outcome CO4 of the Computer Networks (CSA07) syllabus, which requires students to design network topologies — star, mesh, bus, and hybrid — using appropriate networking devices, transmission media, and cabling standards, at Bloom's Taxonomy levels BL2 (Understand) and BL6 (Create). The report below explicitly demonstrates both levels: each scenario begins with an understanding-level analysis of why a topology fits the given constraints, and proceeds to a creation-level design that specifies real devices, real cabling standards (Cat5e, Cat6, RG-58 coaxial, single-mode and multi-mode fibre), and a working network layout.

How to Read This Report

Diagrams are provided for every topology and are drawn to reflect actual physical connections rather than abstract network graphs, so that a reader can trace a cable from an end device back to the core of the network. Tables summarise devices, media, and comparative scores so that the same information is available in both narrative and tabular form, consistent with the documentation and presentation criteria of the grading rubric.

2. Star Topology — 20-Person Startup Office

2.1 Problem Statement and Constraints

A newly funded startup with 20 employees is setting up its first office network. The organisation has no dedicated IT department, expects moderate growth over the next two to three years, and needs a network that is simple to install, easy to troubleshoot, and inexpensive to maintain. Typical office traffic includes web browsing, cloud-based productivity tools, shared printing, and VoIP calling.

2.2 Requirement Analysis

The dominant constraints in this scenario are manageability and fault isolation rather than extreme redundancy. Because the team is small and centrally located on what is presumably a single office floor, a topology that centralises all connections at one point makes both initial installation and ongoing troubleshooting straightforward. The design must also leave headroom for a small number of additional employees, printers, or VoIP handsets without requiring a redesign.

2.3 Proposed Topology: Star

A star topology is the correct fit for this scenario. Every device in the office connects with its own dedicated cable directly to a central switch, rather than sharing a single line with other devices. This physical arrangement gives the network two properties that a startup values highly: first, any single cable or device fault is isolated to that one connection, so if one employee's cable fails, the other 19 employees continue working without interruption; second, centralising all cabling at one point makes it easy to add, move, or reconfigure a workstation later, since only the run between that device and the switch needs to change.

Star Topology — 20-Person Startup Office

All devices connect directly to a central managed switch via Cat6 UTP

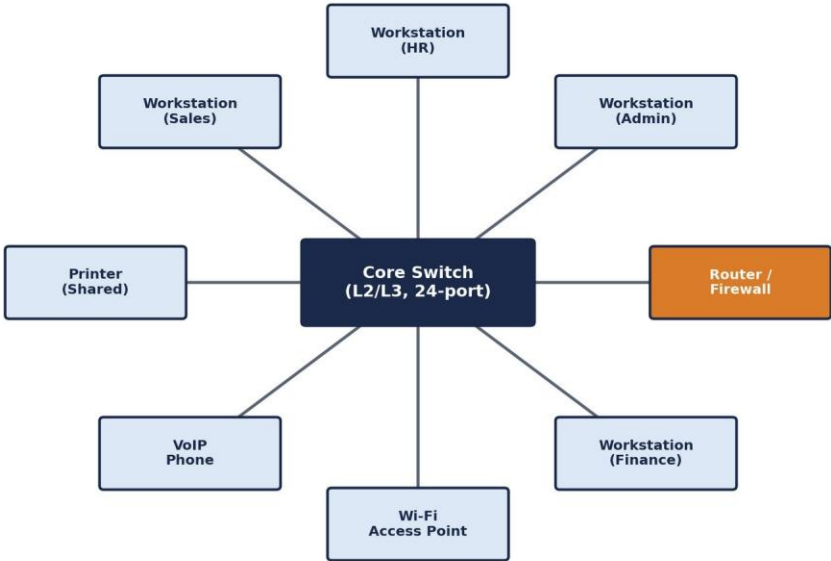


Figure 2.1 — Star topology for a 20-person startup office. All endpoints radiate from a single managed core switch.

2.4 Devices, Media, and Cabling Standard

Component	Specification	Role in the Design
Core switch	Layer 2/3 managed switch, 24-port (e.g. Cisco Catalyst 1300, Netgear ProSafe)	Central connection point for all workstations, printers, and phones; supports VLANs if the company grows
Router / firewall	Cisco RV-series or pfSense appliance	Provides internet access, DHCP, and NAT; enforces basic perimeter security
Wireless access point	802.11ac/ax AP, switch-connected	Extends coverage to laptops and mobile devices without extra cabling
Horizontal cabling	Cat6 UTP, point-to-point	Gigabit-capable up to 100 m per run — more than sufficient for a single office floor
End devices	Workstations, shared printer, VoIP handsets	20 employees plus shared peripherals, each home-run to the switch

2.5 Justification and Trade-offs

The key advantage of a star design is fault isolation: because each device has its own dedicated cable back to the switch, a single cable break, connector fault, or NIC failure affects only that one node. All other

employees remain online, and the fault is trivial to locate — the network administrator simply checks the corresponding switch port. Centralised cable management at the switch also simplifies future moves, adds, and changes, since new drops can be run without touching any existing connection.

The principal trade-off is that the central switch becomes a single point of failure for the network as a whole: if the switch itself fails, every connected device loses connectivity simultaneously. For a 20-person startup this risk is acceptable given the cost of introducing redundancy, but it is explicitly why a bank or hospital — discussed in the next section — cannot rely on a pure star design at its core.

2.6 Testing and Validation Approach

- Verify Gigabit link speed and duplex negotiation on every switch port after installation.
- Confirm DHCP lease assignment and correct VLAN membership (if configured) for each device.
- Simulate a single cable disconnect to confirm that only the affected device loses connectivity.
- Load-test the shared printer and Wi-Fi access point under concurrent use to confirm no bottleneck at the switch uplink.

3. Mesh Topology — Bank Core Network

3.1 Problem Statement and Constraints

A bank operates ATMs, teller terminals, and online banking services that must remain available essentially around the clock. Regulatory requirements typically mandate a documented uptime target (often expressed as 99.99% availability or higher), and any unplanned outage can translate directly into lost transactions, regulatory scrutiny, and reputational damage. The network must therefore be engineered so that no single point of failure can take core banking services offline.

3.2 Requirement Analysis

Unlike the startup scenario, the bank's dominant requirement is fault tolerance, not cost minimisation. Every critical link — between core switches, between the core and the internet edge, and between the core and each branch or data-centre distribution layer — must have at least one redundant alternate path. Latency also matters: financial transactions and interbank settlement traffic are latency-sensitive, which pushes the design towards fibre rather than copper at the core.

3.3 Proposed Topology: Partial Mesh

A full mesh, in which every device connects directly to every other device, gives the theoretical maximum redundancy but becomes impractical to cable and manage as the number of nodes grows, since the number of links grows quadratically. For a bank's core network, a partial mesh is the practical compromise: all core devices — core switches, routers, and firewalls — connect directly to each other, forming a fully meshed core, while the distribution layer (branches and the data centre) connects redundantly to at least two core devices rather than to all of them.

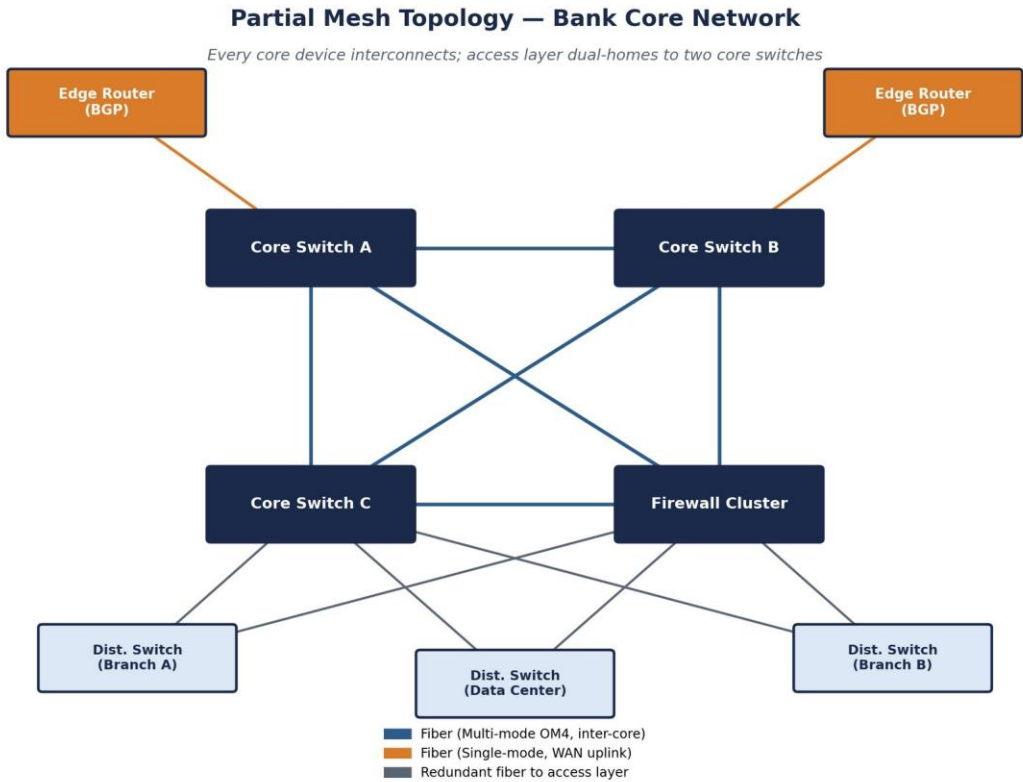


Figure 3.1 — Partial mesh topology at the bank's core, with dual-homed distribution switches and redundant BGPcapable edge routers.

3.4 Devices, Media, and Cabling Standard

Component	Specification	Role in the Design
Core switches / firewall cluster	Enterprise multilayer switches in a fully meshed cluster	Form the resilient core; any single core device or link can fail without isolating the network
Edge routers	Dual enterprise-grade routers (Cisco ASR / Juniper MX), BGP	Redundant internet connectivity with automatic route failover
Inter-core links	Multi-mode OM4 fibre (within data centre)	High bandwidth, immune to electromagnetic interference, low latency
Inter-building links	Single-mode fibre (between branches / buildings)	Long-distance, high-bandwidth backbone connectivity
Distribution switches	Dual-homed to at least two core switches	Branch and data-centre access layers inherit core redundancy
Supporting infrastructure	Redundant power supplies, UPS, RAID storage	Protects against power and storage failure in addition to link failure

3.5 Justification and Trade-offs

The justification for mesh at the core is straightforward: when any single link fails, dynamic routing protocols such as OSPF or BGP recalculate a path around the failure and reroute traffic through an alternative link, typically within milliseconds. Because the distribution layer is dual-homed rather than fully meshed, the bank gets the vast majority of the resilience benefit of a full mesh while keeping the number of physical links, and therefore cost and cabling complexity, manageable.

The trade-offs are cost and complexity: fibre optic cabling, enterprise-grade routers with BGP licensing, and redundant power and storage systems are all significantly more expensive than the equipment used in the startup scenario. This expense is justified because the cost of downtime — in lost transactions, regulatory penalties, and reputational harm — far exceeds the capital cost of the redundant infrastructure.

3.6 Testing and Validation Approach

- Simulate the failure of each core link individually and confirm sub-second reconvergence via OSPF/BGP.
- Measure end-to-end latency across the fibre backbone under peak transaction load.
- Verify automatic failover of the edge routers by disabling one BGP session and confirming the alternate path takes over without manual intervention.
- Conduct a full data-centre power-failure drill to confirm UPS and RAID redundancy hold the network and storage layer online.

4. Bus Topology — Small Classroom Lab

4.1 Problem Statement and Constraints

A school or college needs to network a small computer lab used for supervised activities such as internet browsing, basic file sharing, and coursework. The defining constraint is budget: the institution wants the lowest possible hardware cost and has no requirement for high availability, since the lab is used under supervision during scheduled class periods and a brief outage is a minor inconvenience rather than a business risk.

4.2 Requirement Analysis

Because uptime and scalability are not priorities, the design should minimise the number of active networking components. Traditional bus topology achieves this by eliminating the central switch or hub altogether: every computer taps directly into a single shared backbone cable. This removes the single largest cost item in a star design — the managed switch — at the expense of resilience and performance, both of which are acceptable trade-offs for this environment.

4.3 Proposed Topology: Bus

A single coaxial cable (RG-58, 50-ohm) runs the length of the room and forms the shared backbone, or 'bus'. Each computer connects to this backbone through a T-connector and BNC connector, and 50-ohm

terminators are placed at both physical ends of the cable to absorb the signal and prevent reflection, which would otherwise corrupt data on the line. A more contemporary low-cost variant replaces the coaxial cable and terminators with a single inexpensive unmanaged 8- or 16-port switch and Cat5e UTP patch cables, which achieves the same bus-like simplicity and low cost while being easier to source than legacy coaxial hardware.

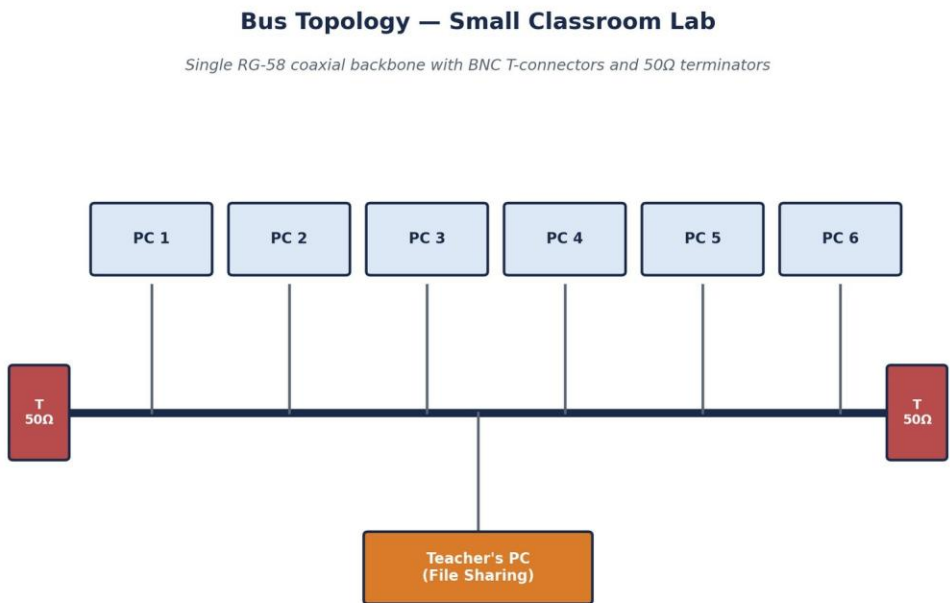


Figure 4.1 — Classic bus topology for a classroom lab, showing the RG-58 backbone, T-connectors, and end terminators.

4.4 Devices, Media, and Cabling Standard

Component	Specification	Role in the Design
Backbone cable	RG-58 coaxial, 50-ohm (or single Cat5e/Cat6 run to an unmanaged hub in the modern variant)	Shared communication line carrying all traffic between computers
T-connectors / BNC connectors	Standard BNC T-piece per workstation	Taps each computer onto the shared backbone
Terminators	50-ohm terminating resistors, one at each end of the cable	Absorb signal energy and prevent reflection/echo on the bus
Unmanaged switch (modern variant)	8- or 16-port unmanaged switch	Low-cost substitute for coaxial hardware; simpler to source and maintain
Teacher's PC	Standard workstation with built-in OS file sharing	Acts as the informal file-sharing point; no dedicated server required

4.5 Justification and Trade-offs

The primary justification for bus topology here is hardware cost: eliminating the central switch or hub removes the single most expensive component of a comparable star design, and coaxial cable together with T-connectors and terminators is inexpensive and simple to install along a single row of desks. No dedicated server is required, since a teacher's PC can share files using built-in operating-system file sharing.

The limitation is that the network is susceptible to collisions as more machines transmit simultaneously, and because every device shares the same physical line, a single cable break or a bad terminator disrupts the entire lab rather than just one machine, unlike the fault isolation offered by a star design. For a supervised classroom environment with light, non-critical usage, this trade-off is entirely acceptable given the cost savings.

4.6 Testing and Validation Approach

- Verify signal integrity end-to-end using a cable tester before connecting any workstation.
- Confirm both terminators are correctly rated (50-ohm) and firmly seated to prevent reflection.
- Test file sharing from the teacher's PC to every workstation under normal class-size load.
- Deliberately disconnect one T-connector to confirm — and document for students — that the entire segment loses connectivity, reinforcing the topology's known limitation.

5. Hybrid Topology — Large University Campus

5.1 Problem Statement and Constraints

A university campus must serve thousands of users spread across lecture halls, laboratories, libraries, dormitories, and administrative buildings. Usage patterns differ sharply by location — labs need highbandwidth wired access, dormitories need dense Wi-Fi coverage, and administrative buildings need secure access to student records — while the campus as a whole must remain easy to expand as new buildings are constructed.

5.2 Requirement Analysis

No single topology from the previous sections satisfies a campus network on its own. A pure mesh would be prohibitively expensive to cable across dozens of buildings; a pure star would create an unmanageable single point of failure at campus scale; and a pure bus could not deliver the bandwidth or reliability that thousands of concurrent users require. The requirement is therefore for a layered design that applies the right topology at each layer of the network rather than one topology for the whole campus.

5.3 Proposed Topology: Hybrid (Three-Tier Hierarchical)

A hybrid topology combines a redundant fibre ring or partial mesh at the campus backbone with a star topology inside each building, and — where appropriate — bus or tree structures within individual labs for cost efficiency. This maps directly onto the industry-standard three-layer hierarchical model: core, distribution, and access. At the campus core, high-speed single-mode fibre (10 Gbps or 40 Gbps) interconnects core routers and multilayer switches in a ring or partial-mesh arrangement, so that no single fibre cut isolates an entire building. Each building has its own distribution switch, connected to the campus core through dual fibre uplinks for redundancy. Within a building, a conventional star topology branches from the distribution switch down to floor-level access switches, which then connect individual devices via Cat6 cabling.

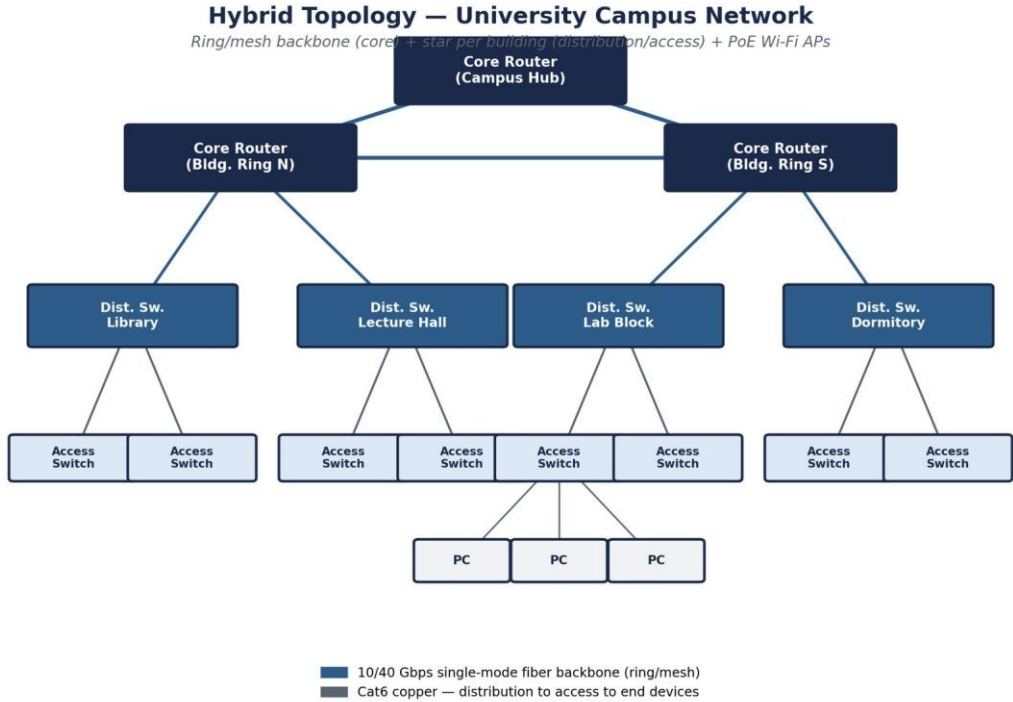


Figure 5.1 — Hybrid three-tier campus network: fibre ring/mesh core, per-building distribution switches, and star-wired access layers.

5.4 Devices, Media, and Cabling Standard

Layer	Devices	Media / Cabling Standard
Core (campus backbone)	Core routers / multilayer switches interconnected in a ring or partial mesh	Single-mode fibre, 10 Gbps or 40 Gbps
Distribution (per building)	One distribution switch per building, dual-uplinked to the core	Single-mode or multi-mode fibre uplinks
Access (per floor / lab)	Floor-level access switches in a star from the distribution switch	Cat6 UTP to individual devices
Wireless	PoE-powered access points connected to access switches	Power over Ethernet — no separate power cabling required

5.5 Justification and Trade-offs

This hierarchical model is the industry standard for enterprise and campus networks precisely because it isolates change: adding a new building means installing one new distribution switch and running fibre to the core, without touching the existing star-wired buildings or any other part of the network. The ring or mesh backbone ensures that a single fibre cut does not isolate an entire building, while the star topology inside each building keeps in-building management simple and keeps fault isolation at the level of a single device, exactly as in the startup scenario. Power over Ethernet for wireless access points removes the need for separate electrical cabling to every AP location, simplifying installation across dozens of buildings.

The trade-off is design and installation complexity: a hybrid network requires careful planning of IP addressing, VLANs, and routing across every layer, and demands more skilled staff to operate than a single flat topology. For an institution the size of a university campus, this complexity is unavoidable and is offset by the scalability and resilience the design provides.

5.6 Testing and Validation Approach

- Verify ring/mesh backbone failover by disabling one fibre link and confirming automatic rerouting.
- Load-test each building's distribution uplink during peak class-change periods.
- Confirm PoE budget is sufficient for all deployed access points per switch.
- Validate that a newly added building can be provisioned without disrupting existing buildings, using a staging/test VLAN before go-live.

6. Multi-Department Company Network with VLAN Segmentation

6.1 Problem Statement and Constraints

A mid-sized company with multiple functional departments — Human Resources, Finance, Sales, and IT — needs a structured, scalable network that lets employees within each department communicate efficiently while keeping departmental traffic logically separated for security and performance reasons. Finance and HR in particular handle sensitive data that should not be broadcast to, or easily reachable from, unrelated departments.

6.2 Requirement Analysis

The core requirement here is logical rather than physical segmentation: departments may sit on the same physical switches and cabling, but their traffic must be isolated at the network layer. This calls for a hierarchical design built around Virtual LANs (VLANs), where each department is assigned its own VLAN, combined with a Layer 3 device that can route between VLANs only where inter-departmental communication is explicitly required.

6.3 Proposed Design: Hierarchical Access/Core with Inter-VLAN Routing

Each department connects through its own dedicated Layer 2 access switch, allowing devices within that department — computers, printers, and other peripherals — to communicate directly with each other. Every departmental switch then uplinks to a central Layer 3 switch or router, which forms the core layer of the design and enables controlled communication between departments through inter-VLAN routing. VLANs are configured so that each department's broadcast domain is logically separated from the others, which both improves security — since a compromised device in one VLAN cannot directly reach devices in another without passing through the routed core, where access control lists can be applied — and reduces broadcast traffic, since broadcasts stay confined to their own VLAN rather than flooding the entire company network. A router with firewall functionality connects the internal network to the internet, and redundant links or backup paths can be added at the core for reliability.

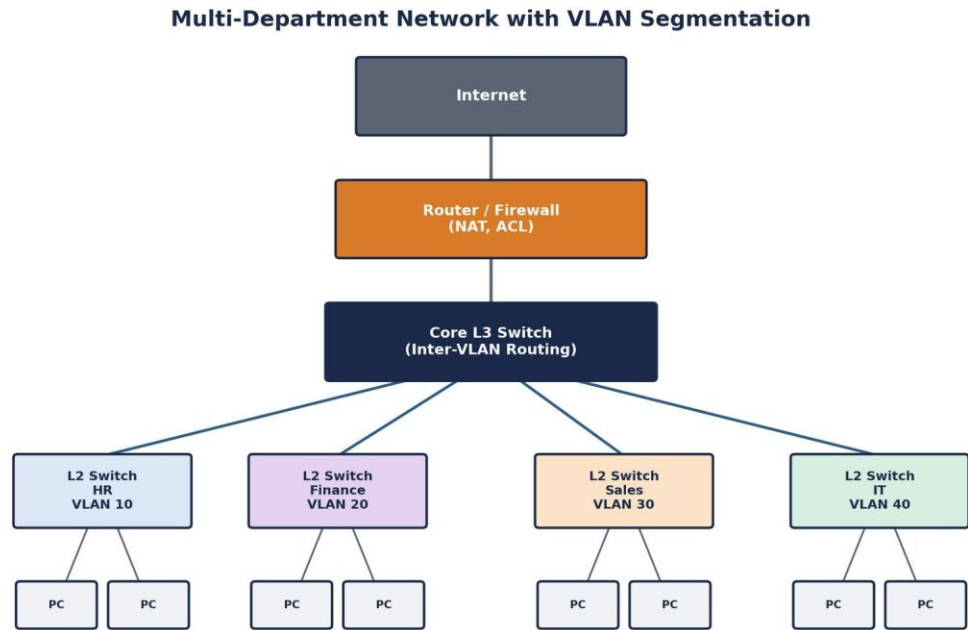


Figure 6.1 — Multi-department network with per-department VLANs and a Layer 3 core performing inter-VLAN routing.

6.4 Devices, Media, and Addressing Plan

Department	VLAN ID	Suggested Subnet	Access Devices
Human Resources	VLAN 10	192.168.10.0/24	L2 access switch + workstations, printer
Finance	VLAN 20	192.168.20.0/24	L2 access switch + workstations, secure file server
Sales	VLAN 30	192.168.30.0/24	L2 access switch + workstations, VoIP phones
IT	VLAN 40	192.168.40.0/24	L2 access switch + workstations, management console
Component	Specification	Role in the Design	
Core Layer 3 switch/router	Multilayer switch with inter-VLAN routing enabled	Routes traffic between department VLANs; central policy enforcement point	
Access switches	Layer 2 managed switches, one per department	Carry each department's VLAN to its end devices	

Edge router/firewall	NAT- and ACL-capable router	Connects the internal VLANs to the
Component	Specification	Role in the Design
		internet with perimeter security
Structured cabling	Cat6 UTP horizontal runs; fibre or link aggregation for switch uplinks	Gigabit connectivity from access layer to core

6.5 Justification and Trade-offs

This design improves performance, because broadcast traffic is confined within each VLAN rather than flooding the whole company; it enhances security, because access control lists on the Layer 3 core can restrict exactly which departments are permitted to reach which resources — for example, allowing Finance to reach a shared file server while denying Sales direct access to HR systems; and it allows easy expansion, since adding a new department is simply a matter of provisioning a new VLAN and access switch and adding one routed interface at the core.

The trade-off is that inter-VLAN routing and access-control configuration require more networking expertise to design and maintain correctly than a flat, unsegmented network, and a misconfigured ACL can either over-restrict legitimate traffic or under-restrict sensitive traffic. Careful IP addressing and subnetting, as summarised in the table above, together with documented routing and firewall policies, mitigate this risk.

6.6 Testing and Validation Approach

- Confirm each department's devices receive addresses only within their assigned VLAN subnet.
- Verify inter-VLAN ACLs by attempting both permitted and denied cross-department connections.
- Measure broadcast traffic within a VLAN to confirm it does not propagate to other VLANs.
- Test failover of redundant core links, where configured, under simulated link failure.

7. Comparative Analysis of the Five Designs

Although each scenario in this report calls for a different topology, they share a common design methodology: gather requirements, analyse traffic and growth, select a topology, choose devices and media, plan addressing, implement redundancy and security, test, and document. The flowchart below summarises this methodology, which was applied — implicitly or explicitly — in every section above.

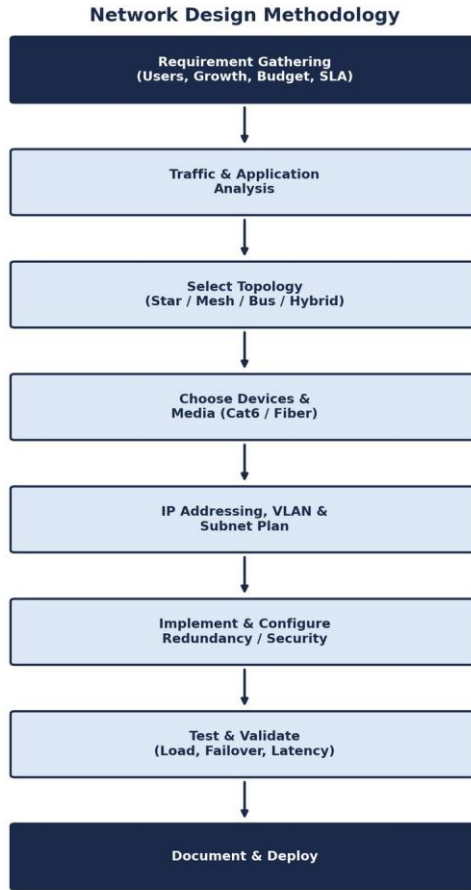


Figure 7.1 — General network design methodology applied across all five scenarios.

7.1 Side-by-Side Comparison

Criterion	Star (Startup)	Mesh (Bank)	Bus (Classroom)	Hybrid (Campus)	VLAN Core (Company)
Relative cost	Medium	High	Very low	High	Medium-high
Fault tolerance	Low (single switch)	Very high	Very low	Very high	High (with redundant core)
Scalability	Medium	Medium	Very low	Very high	Very high
Ease of management	High	Medium (skilled staff)	Medium	Medium	Medium
Criterion	Star (Startup)	Mesh (Bank)	Bus (Classroom)	Hybrid (Campus)	VLAN Core (Company)
Primary media	Cat6 UTP	Single/multimode fibre	RG-58 coax / Cat5e	Fibre core + Cat6 access	Cat6 + fibre uplinks

Best suited for	Small single-site office	Mission-critical finance	Budget-limited lab	Multi-building campus	Multi-department enterprise
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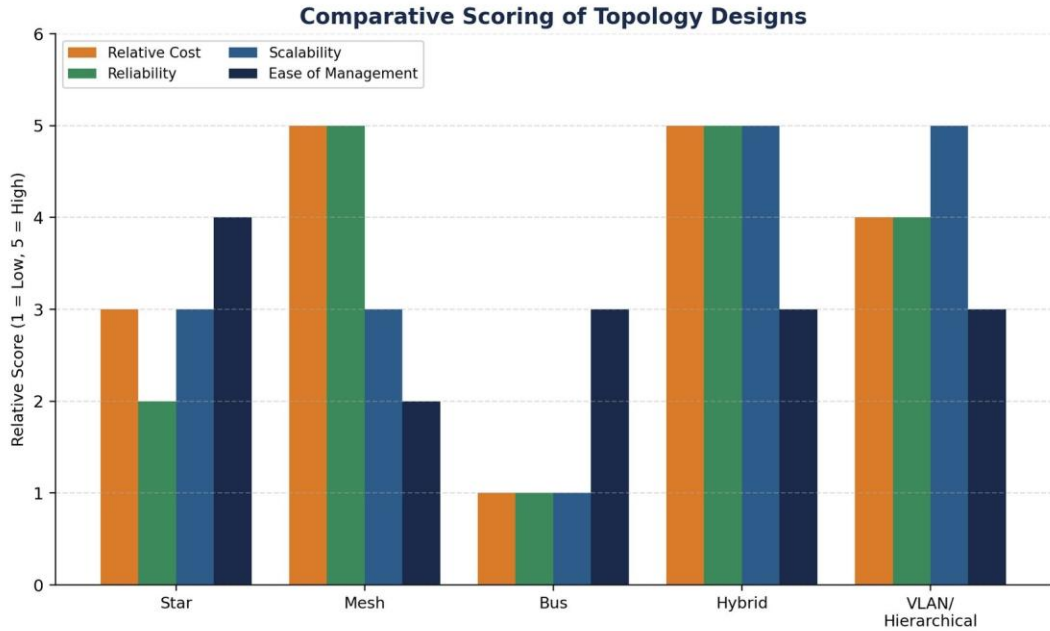


Figure 7.2 — Relative scoring of each design across cost, reliability, scalability, and ease of management (1 = low, 5 = high).

7.2 Key Insight

The comparison makes the underlying principle of this report explicit: cost, fault tolerance, and scalability trade off against one another, and the correct topology is the one whose trade-offs match the operating context. The bus topology scores lowest on almost every technical criterion but is nonetheless the correct choice for the classroom lab, because its one strength — minimal cost — is precisely what that scenario requires. Conversely, the mesh and hybrid designs score highest on reliability and scalability but would be an unjustifiable expense for a 20-person startup or a single classroom.

8. Alignment with Assessment Rubric

The table below maps each rubric criterion from the Annexure A grading scheme to the specific section(s) of this report where that criterion is addressed, to make the basis for evaluation transparent.

Rubric Criterion	Weightage	Addressed In
Problem Analysis	15%	Sections 2.1–2.2, 3.1–3.2, 4.1–4.2, 5.1–5.2, 6.1–6.2 (Problem Statement and Requirement Analysis in every scenario)
Solution Design	20%	Sections 2.3, 3.3, 4.3, 5.3, 6.3 (Proposed Topology / Design in every scenario)

Technical Implementation	20%	Device, media, and cabling tables (2.4, 3.4, 4.4, 5.4, 6.4) with named products and standards
Use of Tools & Technologies	10%	Specific device models, cabling standards (Cat5e/Cat6/RG-58/fibre), routing protocols (OSPF/BGP), and VLAN/subnet planning
Problem-Solving Approach	10%	Section 7.1 general design methodology flowchart applied consistently across all scenarios
Innovation & Creativity	10%	Hybrid three-tier campus design (Section 5) and VLAN-segmented enterprise core (Section 6)
Testing & Validation	5%	Testing and Validation Approach subsection at the end of every scenario, plus Section 9
Documentation & Presentation	5%	Structured headings, labelled diagrams, comparison tables, and table of contents throughout
Teamwork / Collaboration	5%	Declaration page; applicable where the task is completed as a team submission

9. Common Testing and Validation Methodology

While each scenario above includes design-specific tests, a shared validation methodology applies across all five networks before they are considered production-ready. This methodology is organised into four stages.

9.1 Connectivity Verification

- Confirm physical link status (link light / interface up) on every port before proceeding to logical tests.
- Verify correct IP addressing, subnet mask, and default gateway on a sample of end devices in each segment.
- Use ping and traceroute to confirm reachability within a segment and across routed boundaries where permitted.

9.2 Performance Testing

- Measure throughput between representative endpoints under normal and peak load using an iperf-style tool.
- Check for packet loss and jitter on VoIP or latency-sensitive traffic paths.
- Confirm that switch and router CPU/memory utilisation remains within safe operating margins under peak load.

9.3 Redundancy and Failover Testing

- For mesh and hybrid designs, deliberately disable one redundant link at a time and confirm automatic rerouting with minimal packet loss.
- For star and bus designs, disconnect a single end-device cable and confirm the fault remains isolated to that device (star) or is correctly documented as a shared-segment risk (bus).
- Test UPS and backup power failover where redundant power is part of the design.

9.4 Security Validation

- Confirm VLAN separation by attempting communication across VLANs without an explicit permit rule and verifying it is blocked.
- Review firewall and ACL rules against the intended access policy for each department or zone.
- Confirm default credentials have been changed on all managed devices and that management access is restricted to authorised staff.

10. Transmission Media and Cabling Standards — Detailed Reference

This section consolidates the transmission media and cabling standards referenced across all five designs into a single technical reference, since the choice of medium has a direct effect on achievable bandwidth, maximum run length, and susceptibility to interference — all of which influenced the topology decisions made earlier in this report.

10.1 Copper Media

Standard	Typical Use	Max. Segment Length	Typical Bandwidth
Cat5e UTP	Legacy/budget horizontal cabling (classroom lab variant)	100 m	Up to 1 Gbps
Cat6 UTP	Standard horizontal cabling (startup, campus access, department access)	100 m	Up to 10 Gbps (55 m for 10GBASE-T)
RG-58 coaxial (50-ohm)	Legacy bus backbone (classroom lab)	185 m (10BASE2 segment)	10 Mbps

10.2 Fibre Optic Media

Standard	Typical Use	Max. Distance	Key Advantage
Multi-mode OM4 fibre	Intra-data-centre core links (bank)	Up to 550 m at 10 Gbps	High bandwidth over short/medium distances

Single-mode fibre	Inter-building/campus backbone (bank, university)	Multiple kilometres	Very high bandwidth over long distances; immune to EMI
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10.3 Why Media Choice Follows Topology, Not the Reverse

A recurring theme across this report is that the transmission medium is selected to serve the topology's purpose rather than the other way around. The classroom lab's bus topology is only economically sensible with low-cost coaxial or basic UTP cabling; if the same lab were wired with single-mode fibre, the cost savings that justify a bus topology in the first place would disappear. Conversely, the bank's mesh core would be pointless if built on copper, since electromagnetic interference and distance limitations would undermine the very reliability the mesh design is meant to provide. Matching medium to topology, and topology to organisational requirement, is therefore a single continuous design decision rather than three separate ones.

11. Security and Reliability Considerations Across Designs

Beyond topology and cabling, every design in this report depends on a small set of shared security and reliability practices. This section draws these together explicitly, since documentation and presentation quality benefit from stating cross-cutting concerns once rather than repeating them in each scenario.

11.1 Perimeter and Access Security

- All five designs place a router or firewall between the internal network and the internet, performing NAT and, where relevant, stateful packet filtering.
- The department-VLAN design (Section 6) additionally applies access control lists at the Layer 3 core to restrict which VLANs may reach which resources.
- The bank design (Section 3) layers a dedicated firewall cluster at the core in addition to the edge routers, reflecting the higher sensitivity of financial data.

11.2 Physical and Power Redundancy

- The bank and university designs both specify redundant power supplies and, for the bank, UPS and RAID storage, since a power interruption is as damaging to availability as a network fault.
- The startup and classroom designs intentionally omit these measures, since the cost of redundant power is not justified by the risk profile of those environments.

11.3 Change Management

Each design's fault-isolation properties directly influence how safely changes can be made. Star, hybrid, and VLAN-segmented designs allow a single device, floor, or department to be modified, patched, or expanded without affecting unrelated parts of the network — the change is contained by the topology itself. The bus design offers no such containment: any change to the shared backbone risks disrupting the

whole segment, which is an accepted limitation given the classroom's low-risk usage profile but would be unacceptable in any of the other four scenarios.

12. Conclusion

This report has examined five distinct industry scenarios and, for each, proposed a network topology grounded in the specific operational constraints of that environment: a star topology for a small startup office where manageability and fault isolation matter most; a partial mesh at the core of a bank's network where near-zero downtime is a regulatory and business necessity; a bus topology for a budget-constrained classroom lab where minimising hardware cost outweighs the need for redundancy; a hybrid three-tier design for a university campus that must scale across many buildings while remaining manageable; and a VLAN-segmented hierarchical design for a multi-department company that needs logical separation and controlled inter-departmental routing.

Across all five designs, a consistent methodology was applied: understand the requirement, analyse the traffic and growth pattern, select the topology whose strengths match the constraint that matters most, specify concrete devices and cabling standards, and validate the result through connectivity, performance, redundancy, and security testing. The comparative analysis in Section 7 makes explicit what each scenario demonstrates individually — that network design is fundamentally an exercise in matching trade-offs to context, and that no topology is 'best' in the abstract, only best for a given set of constraints.

This exercise directly satisfies Course Outcome CO4 of the Computer Networks (CSA07) syllabus, demonstrating both an understanding of star, mesh, bus, and hybrid topologies (BL2) and the ability to create concrete, justified network designs using real networking devices, transmission media, and cabling standards (BL6).

13. References

1. Annexure A — Industry Problem Solving Task, Computer Networks (CSA07), SIMATS Engineering, Assessment Tool 2.
2. Cisco Systems — Networking Academy documentation on switching, routing, and VLAN configuration.
3. TIA/EIA-568 Commercial Building Telecommunications Cabling Standard (structured cabling, Cat5e/Cat6).
4. IEEE 802.3 Ethernet Standards (physical and data-link layer specifications for copper and fibre media).
5. IEEE 802.1Q — Standard for VLAN tagging and inter-VLAN routing.
6. Kurose, J. F., & Ross, K. W. — Computer Networking: A Top-Down Approach, Pearson.
7. Tanenbaum, A. S., & Wetherall, D. J. — Computer Networks, Pearson.